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Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA1506-CLOUD COMPUTING AND BIG DATA ANALYTICS

THE ROLE OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING IN  
CLOUD PLATFORMS

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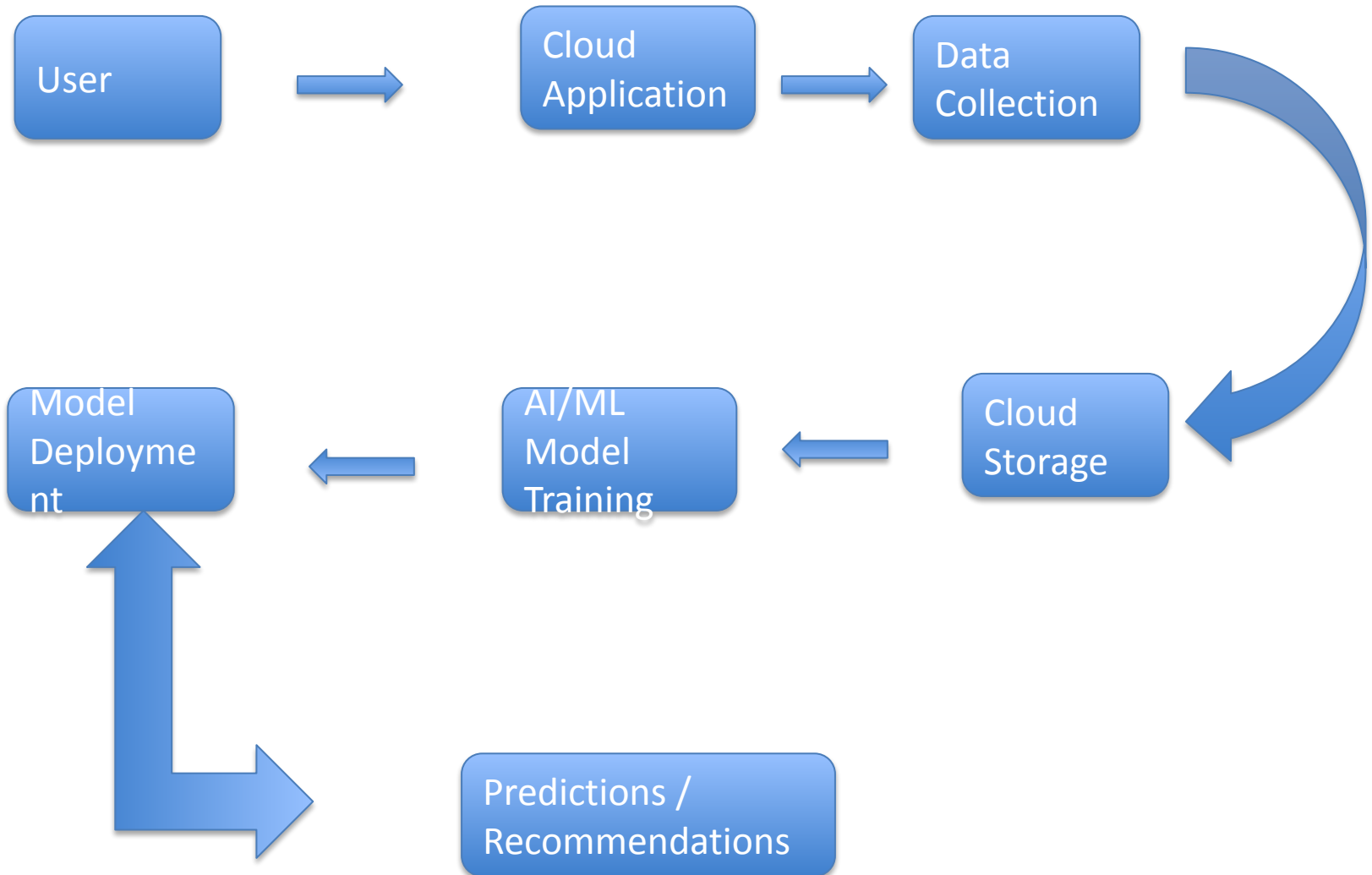
# Introduction

- Artificial Intelligence (AI) enables computers to perform tasks that normally require human intelligence.
- Machine Learning (ML) is a branch of AI that learns from data and improves automatically.
- Cloud platforms provide scalable computing, storage, and networking services.
- AI and ML in the cloud make advanced technologies accessible to everyone.
- Organizations use cloud-based AI to reduce costs and improve productivity.

# What is Artificial Intelligence?

- AI enables machines to mimic human thinking and decision-making.
- It performs tasks like speech recognition and image processing.
- AI systems analyze large amounts of data quickly.
- AI reduces manual effort and improves accuracy.
- It is widely used in healthcare, banking, education, and transportation.

# Architecture / Process Diagram



# What is Machine Learning?

- ML is a subset of AI.
- ML algorithms learn patterns from historical data.
- Models improve their performance through training.
- ML predicts future outcomes using existing data.
- It supports automation in various industries.

# What are Cloud Platforms?

- Cloud platforms deliver computing resources over the Internet.
- Users pay only for the services they use.
- Cloud services are available anytime and anywhere.
- They provide storage, processing, networking, and security.
- Cloud platforms support AI and ML model development.
- **Popular Cloud Platforms**
  - Amazon Web Services (AWS)
  - Microsoft Azure
  - Google Cloud Platform (GCP)
  - IBM Cloud

# AI and ML Services in Cloud Platforms

- Pre-trained AI models
- Machine Learning model training
- Image and video recognition
- Natural Language Processing (NLP)
- Speech recognition and translation

# Future Scope

- Generative AI in cloud computing
- Edge AI integration
- Automated Machine Learning (AutoML)
- Intelligent cloud security
- AI-powered business analytics
- Smart cities and IoT integration

# Conclusion

- AI and ML are transforming cloud computing.
- Cloud platforms make AI technologies affordable and scalable.
- Businesses gain faster insights and better decision-making.
- AI-powered cloud services improve efficiency across industries.
- The future of cloud computing will rely heavily on intelligent AI solutions.

THANK YOU